P802.1AXdz/D1.1
February 7, 2025
(Amendment to IEEE Std 802.1AX-2020)

Draft Standard for Local and metropolitan area networks—

Link Aggregation

Amendment 1: YANG for Link Aggregation

Prepared by the
Time-Sensitive Networking (TSN) Task Group of IEEE 802.1

Sponsor
LAN/MAN Standards Committee
of the
IEEE Computer Society

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The text proper of this draft begins with the Title page.

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The IEEE SA PAR (Project Authorization Request) and CSD (Criteria for Standards Development established by IEEE 802) are summarized in these cover pages and links are provided to the full text of both PAR and CSD. As part of the IEEE 802® process, the text of the PAR and CSD of each project is reviewed regularly to ensure their continued validity. A vote of "Approve" on this draft is also an affirmation that the PAR and CSD for this project are still valid.

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Comments on this draft may be sent to the 802.1 email exploder, to the Editors, or to the Chairs of the 802.1 Working Group and Time-Sensitive Networking (TSN) Task Group.

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PAR (Project Authorization Request) and CSD

This page is a draft, based on the proposed PAR and CSD as of the close of the May 2023 802.1 Interim Meeting.

Extracts from the PAR, as approved by IEEE NesCom February 15, 2024:

https://development.standards.ieee.org/myproject-web/public/view.html#pardetail/10885

and the CSD (Criteria for Standards Development):

https://mentor.ieee.org/802-ec/dcn/23/ec-23-0238-00-ACSD-p802-1axdz.pdf

follow.

Scope of the project:

This amendment specifies YANG modules that allows configuration and status reporting for systems implementing Link Aggregation, and optionally Distributed Resilient Network Interconnect, based on the capabilities currently specified in clause 7 (management) and Annex D (Management Information Base definitions). This amendment also includes technical and editorial corrections in the description of existing IEEE Std 802.1AX functionality.

PAR Need for the Project:

YANG (IETF RFC 7950) is a formalized data modeling language that is widely accepted and can be used to simplify network configuration. The ability to manage Link Aggregation via YANG modules is needed for compatibility with modern network management systems.

CSD broad market potential [extract]:

The proposed amendment will support the use of YANG, which has broad industry support in networks that use IEEE Std 802.1AX. Both IEEE Std 802.1AX and YANG are already supported and used by multiple vendors, network providers, and network users. There is a wide interest in the industry to manage Link Aggregation via YANG.

Economic feasibility [extract]:

a) Management using YANG utilizes a balance between end station and infrastructure capabilities; the balance will be similar to that for existing management methods.
b) The cost factors will be similar to those of existing management methods.
c) This project adds YANG capabilities to IEEE Std 802.1AX as a step towards a complete YANG management solution. This helps to eliminate multiple management platforms, thus reduces installation cost.
d) This project adds YANG capabilities to IEEE Std 802.1AX as a step towards a complete YANG management solution. This helps to eliminate multiple management platforms, thus reduces operational cost.

Draft development

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MIB and YANG modules

The MIB and YANG modules that are modified or added by this amendment are attached to the draft pdf as plain text (UTF-8) .mib and .yang files. When a roll up of the current base standard plus this amendment is made available, all the MIB and YANG modules for the roll up are attached.

Introduction to the current draft [1]

This introduction is not part of the draft, and should not be the subject of ballot comments.

D1.1

Incorporates the resolution of comments received during the Working Group ballot of draft 1.0. Significant changes resulting from the ballot include:

— Changing the representation of the 4096 bit mask in the drni-mask grouping from binary to a string listing individual CIDs, or ranges of CIDs, corresponding to bits in the mask that are set to 1.

— Removing several 4096 bit masks from the YANG model because they could be calculated from other information in the model;

— Moving the actor-admin-state leaf from the aggport container to the key-group container.

D1.0

Incorporates the resolution of comments received during the Task Group ballot of draft 0.2. Significant changes resulting from the ballot include:

— Combining the old figures 10-2 and 10-3 into a single page UML-like diagram for Link Aggregation (excluding DRNI);

— Clean up separation of Link Aggregation from the optional DRNI by adding a separate line in the conformance clause options for DRNI, and moving all DRNI related types from *ieee802-dot1ax-types.yang* to *ieee802-dot1ax-drni.yang*;

— Moving the system and system-priority leafs from key-groups to a separate list of aggregation systems. This allows the system mac address to be provided by the system rather than requiring it to be explicitly specified in a configuration model.

Also changed was adding a base identity choice for the admin-conv-link-map and admin-conv-service-map of CSCD. This allows a vendor to augment the yang model with an identity statement specifying pre-defined contents of the table to provide a shorthand for configuration that otherwise could potentially take thousands of lines of configuration.

D0.2

Incorporates the resolution of comments received during the Task Group ballot of draft 0.1.

D0.1

This is an initial draft and comments are requested on all aspects of the draft. It includes a number of notes that may be of help to the Editors as well as informing the initial review process.

Stephen Haddock, 802.1AXdz Editor

[1] The whole or parts of the introduction, possibly updated, to past drafts may be retained at the Editor's discretion, with the most recent introduction first. The introduction to each draft may solicit input on specific subjects.

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Abstract: This amendment to IEEE Std 802.1AX-2020 specifies a Unified Modeling Language (UML)-based model and YANG modules for Link Aggregation configuration and status reporting.

Keywords: Aggregated Link, Aggregator, Distributed Resilient Network Interconnect, DRNI, interconnect, Link Aggregation, Link Aggregation Group, local area network, management, Network-Network Interface, NNI, YANG.

The Institute of Electrical and Electronics Engineers, Inc.
3 Park Avenue, New York, NY 10016-5997, USA

PDF: ISBN 978-X-XXX-XXX-X STDXXXXX
Print: ISBN 978-X-XXX-XXX-X STDPDXXXXX

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<<The following lists will be updated in the usual way prior to publication>>

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<<TBA>>

The following members of the individual balloting committee voted on this standard. Balloters may have voted for approval, disapproval, or abstention.

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When the IEEE-SA Standards Board approved this standard on XX Month 20xx, it had the following membership:

<<TBA>>

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*Member Emeritus

Introduction

This introduction is not part of IEEE Std 802.1AXdz™-20XX, IEEE Standard for Local and metropolitan area networks— Link Aggregation—Amendment 1: YANG for Link Aggregation.

IEEE Std 802.1AXdz™-202X: YANG for Link Aggregation specifies a Unified Modeling Language (UML)-based model and YANG modules for Link Aggregation configuration and status reporting

This standard contains state-of-the-art material. The area covered by this standard is undergoing evolution. Revisions are anticipated within the next few years to clarify existing material, to correct possible errors, and to incorporate new related material. Information on the current revision state of this and other IEEE 802 standards may be obtained from

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USA

Contents

2. Normative references 18

4. Acronyms and abbreviations 19

5. Conformance 20

5.3.2 Link Aggregation options 20

10. YANG Data Model 21

10.1 YANG Framework 21
10.2 Information Model for Link Aggregation Management 21
10.2.1 Link Aggregation model 22
10.2.2 DRNI model 24
10.3 Structure of the Link Aggregation YANG Model 25
10.4 Relationship to other YANG modules 25
10.4.1 IEEE 802.1AX Types Module 25
10.4.2 IETF YANG Types Module 25
10.4.3 IETF Interfaces YANG Module 25
10.4.4 IEEE 802 Types Module 25
10.5 Security Considerations 26
10.5.1 Security considerations of the ieee802-dot1ax YANG modules 26
10.6 YANG schema tree definitions, ... 27
10.6.1 Schema for the ieee802-dot1ax-linkagg YANG module 27
10.6.2 Schema for the ieee802-dot1ax-drni YANG module 28
10.7 YANG modules 29
10.7.1 The ieee802-dot1ax-types YANG module 29
10.7.2 The ieee802-dot1ax-linkagg YANG module 35
10.7.3 The ieee802-dot1ax-drni YANG module 48

Annex A 58
(normative) 58

Protocol implementation conformance statement (PICS) proforma 58

A.2.1 Major capabilities/options 58

Annex G 59
(informative) 59

Bibliography 59

Figures

Figure 10-1 YANG root hierarchy with Link Aggregation YANG modules 22
Figure 10-2 Link Aggregation Configuration and Monitoring Objects 23
Figure 10-3 DRNI Configuration and Monitoring Objects 24

Tables

Table 10-1 Structure of the YANG modules 25

2. Normative references

Insert the following references into Clause 2 in alphanumeric order:

IETF RFC 6241, Network Configuration Protocol (NETCONF), June 2011.

IETF RFC 6991, Common YANG Data Types, July 2013

IETF RFC 7950, The YANG 1.1 Data Modeling Language, August 2016.

IETF RFC 8343, A YANG Data Model for Interface Management, March 2018

IETF RFC 8349, A YANG Data Model for Routing Management (NMDA Version), March 2018

4. Acronyms and abbreviations

Insert the following acronyms into Clause 4 in alphabetical order:

NETCONF	Network Configuration Protocol
UML®	Unified Modeling Language™[1]

[1] UML® is a registered trademark of the Object Management Group, Inc., and Unified Modeling Language™ is a trademark of the Object Management Group, Inc.

5. Conformance

Insert the following text (items j and k) after item i) in the lettered list of 5.3.2

5.3.2 Link Aggregation options

j) Support YANG modules for the management of Link Aggregation capabilities (Clause 10).

k) Support YANG modules for the management of DRNI capabilities (Clause 10).

Insert Clause 10 after Clause 9 as follows:

10. YANG Data Model

This clause specifies the YANG data model comprised of YANG modules that provide control and status monitoring of systems and system components that implement functionality specified in this standard.

This clause:

a) Introduces the YANG framework that governs the naming and hierarchy of configuration and operational data structures in the data models, and the modeling of network interfaces (10.1).

b) Describes the information data model and its relationship to the operational processes and managed objects specified in the other clauses of this standard, and provides a representation, similar to the Unified Modeling Language (UML), of relevant objects (10.2).

c) Describes the structure of the YANG data model, which comprises three YANG modules (10.3).

d) Includes a relationship description of other modules imported in YANG modules (10.4)

e) Reviews security considerations applicable to each of the modules, with specific reference to data nodes in the YANG modules that compose the YANG data model (10.5).

f) Includes each of the YANG modules and its data schema (10.6).

10.1 YANG Framework

The YANG framework applies hierarchy in the following areas:

a) The uniform resource name (URN), as specified in 802d.

b) The YANG objects form a hierarchy of configuration and operational data structures that specify the YANG model.

10.2 Information Model for Link Aggregation Management

The YANG objects are based on the managed objects in Clause 7. A UML-like representation of the management model is provided in the following subclauses.

NOTE—OMG® UMI.2.5[2] [B11] conventions together with C++ language constructs are used in this clause as a representation to convey model structure and relationships..

The purpose of an UML-like[3] diagram is to express the model design on a single piece of paper. The structure of the UML-like representation shows the name of the object followed by a list of properties for the object. The properties indicate its type and accessibility. The UML-like representation is meant to express simplified semantics for the properties. It is not meant to provide the specific datatype as used to encode the object in either MIB or YANG. In the UML-like representation, a box with a white background represents information that comes from sources outside of the IEEE. A box with a gray background represents objects that are specified by this IEEE Standard.

The YANG hierarchical structure that incorporates the Link Aggregation YANG modules supported by this standard is represented by Figure 10-1. In the figures in this clause, items that are shaded gray are described in this document, items with no background shading are specified elsewhere. The YANG data model is

[2]OMG® is a registered trademark of the Object Management Group, Inc.

[3]A description of the UML-like diagrams used in this clause is provided at https://1.ieee802.org/uml-like-diagrams.

realized in three YANG modules. One module *ieee802-dot1ax-types* provides data types that are needed by the Link Aggregation configuration and monitoring objects. The *ieee802-dot1ax-linkagg* module provides the Link Aggregation, Aggregation Port, and Aggregator configuration and monitoring objects. The *ieee802-dot1ax-drni* module provides the Distributed Resilient Network Interface (DRNI) configuration and monitoring objects. The Link Aggregation and DRNI capabilities are not only applicable to IEEE Std 802.1Q bridges, but also (for example) end stations, and routers.

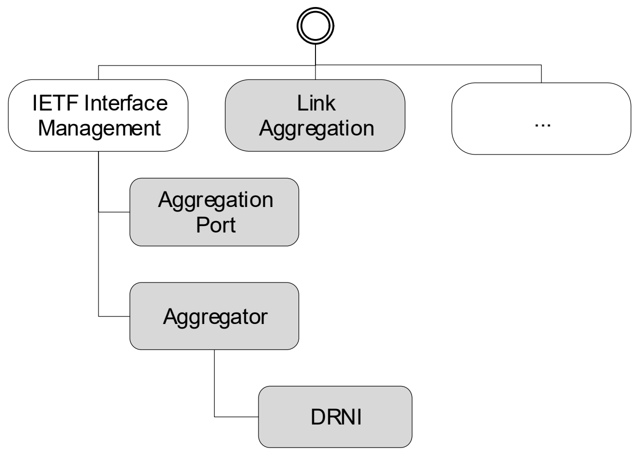

Figure 10-1—YANG root hierarchy with Link Aggregation YANG modules

10.2.1 Link Aggregation model

The Link Aggregation Configuration and Monitoring Objects in Figure 10-2 show the objects that are applicable on a system supporting link aggregation, and the interfaces supporting link aggregation.

The objects on the system consist of a of a list of aggregation systems, and a list of key groups. A key group is the set of aggregators and aggregation ports that share the same system priority, system mac address, and aggregation key, and therefore can potentially form a Link Aggregation Group. Each key group has a unique combination of actor-admin-key and actor-system, and includes parameters that have the same value for any aggregator and/or aggregation port that have the same actor-admin-key and actor-system.

NOTE—See 6.3.5 and 6.3.6 for a description of the identification of Aggregation Ports that can potentially form a Link Aggregation Group, and Figure 6-15 for examples.

Each Aggregation Port is an interface (e.g. of type ethernetCsmacd) augmented with an “aggport” presence container as shown on the right side of Figure 10-2. Each Aggregator is an interface to the entire Link Aggregation Group (typically using type ieee8023adlag) augmented with a “lag” container as shown on the left side of Figure 10-2. The binding of Aggregation Ports to Aggregators is a dynamic function of the Link Aggregation Control Protocol. The Aggregation Ports attached to an Aggregator can be read from the *lower-layer-if* attribute of the Aggregator interface. The Aggregator to which an Aggregation Port is attached can be read through the *higher-layer-if* attribute of the Aggregation Port interface.

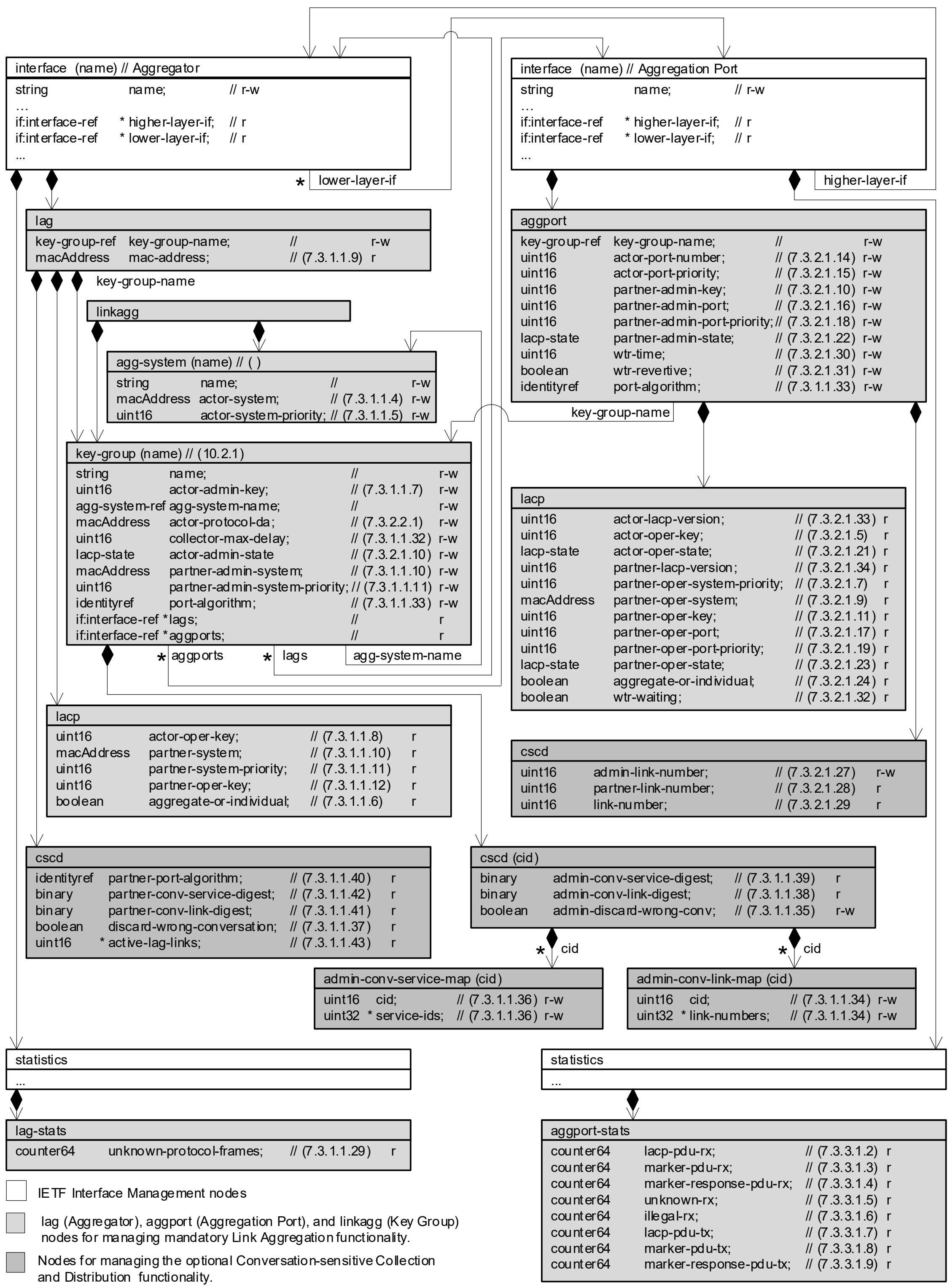

Figure 10-2—Link Aggregation Configuration and Monitoring Objects

10.2.2 DRNI model

The DRNI Configuration and Monitoring Objects in Figure 10-3 show the objects that are applicable to an Aggregator augmented with a “drni” container.

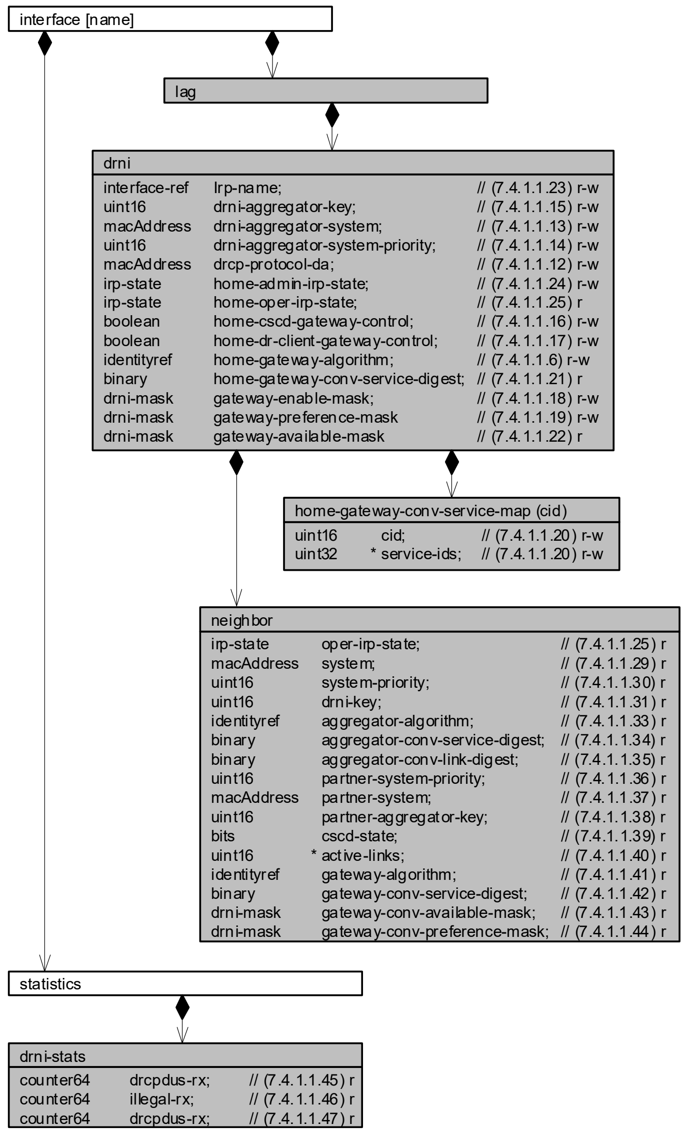

Figure 10-3—DRNI Configuration and Monitoring Objects

10.3 Structure of the Link Aggregation YANG Model

The YANG data model specified in this standard is comprised of three YANG modules. A summary of the modules contained in this clause is represented in Table 10-1.

Table 10-1—Structure of the YANG modules

Module	Subclause	Notes
ieee802-dot1ax-types	10.7.1	Type definitions used for Link Aggregation YANG.
ieee802-dot1ax-linkagg	10.7.2	Link Aggregation Management
ieee802-dot1ax-drni	10.7.3	DRNI Management

In the YANG modules below, if any discrepancy between the DESCRIPTION text and the corresponding specification in any other part of this standard occurs, the specifications outside this clause take precedence.

10.4 Relationship to other YANG modules

This clause describes how the *ieee802-dot1ax-linkagg* and *ieee802-dot1ax-drni* YANG modules are related to the YANG modules that are imported.

10.4.1 IEEE 802.1AX Types Module

The *ieee802-dot1ax-types* module provides reusable types that are used by the *ieee802-dot1ax-linkagg* and *ieee802-dot1ax-drni* modules.

10.4.2 IETF YANG Types Module

The *ietf-yang-types* YANG module (IETF RFC 6991) contains a set of derived YANG types. This document leverages counter64.

10.4.3 IETF Interfaces YANG Module

The *ietf-interfaces* YANG module (IETF RFC 8343) contains a set of YANG definitions for managing network interfaces. This document augments an ietf-interfaces:interface with aggregation-port or aggregator data nodes..

10.4.4 IEEE 802 Types Module

The *ieee802-types* module provides reusable types that are used in IEEE 802 standards.

NOTE—The type for mac-addresses defined in *ieee802-types* has a pattern that allows upper and lower case letters. To avoid issues with string comparison, it is suggested to only use upper case for the letters in the hexadecimal numbers. Implementers using code comparing MAC addresses are advised that there is still an issue with a difference between the IETF mac-address definition and the IEEE mac-address definition.

10.5 Security Considerations

The YANG modules specified in this clause are designed to be accessed via a network configuration protocol (e.g., NETCONF protocol). In the case of NETCONF, the lowest NETCONF layer is the secure transport layer and the mandatory to implement secure transport is SSH. The NETCONF access control model provides the means to restrict access for particular NETCONF users to a pre-configured subset of all available NETCONF protocol operations and content.

It is the responsibility of a system's implementor and administrator to ensure that the protocol entities in the system that support NETCONF, and any other remote configuration protocols that make use of these YANG modules, are properly configured to allow access only to those users who have legitimate rights to read or write data nodes. This standard does not specify how the credentials of those users are to be stored or validated.

10.5.1 Security considerations of the *ieee802-dot1ax* YANG modules

There are several management objects specified in the *ieee802-dot1ax-linkagg* and *ieee802-dot1ax-drni* YANG modules that are configurable (i.e., read-write) and/or operational (i.e., read-only). Such objects could be considered sensitive or vulnerable in some network environments. A network configuration protocol, such as NETCONF (IETF RFC 6241), can support protocol operations that can edit or delete YANG module configuration data (e.g., edit-config, delete-config, copy-config). If this is done in a non-secure environment without proper protection, then negative effects on the network operation are possible.

The following containers, and the objects in these containers, of the *ieee802-dot1ax-linkagg* and *ieee802-dot1ax-drni* YANG modules can be manipulated to interfere with the operation of the Link Aggregation Control Protocol (LACP) or the Distributed Relay Control Protocol (DRCP). This could, for example, cause network instability and result in the loss of service for a large number of end users.

dot1ax-linkagg/link-aggregation/key-group
dot1ax-linkagg/link-aggregation/cscd
dot1ax-linkagg/link-aggregation/cscd/admin-conv-service-map
dot1ax-linkagg/link-aggregation/cscd/admin-conv-link-map
dot1ax-linkagg/aggregation-port
dot1ax-linkagg/aggregation-port/cscd/admin-link-number
dot1ax-linkagg/aggregator
dot1ax-drni/aggregator/drni
dot1ax-drni/aggregator/drni/home-admin-conv-service-map

Some of the readable data in this YANG module could be considered sensitive or vulnerable in some network environments. It is important to control all types of access (e.g., including NETCONF get, get-config operations) to these objects and possibly to even encrypt the values of these objects when sending them over the network. For example, the system name and other information about the remote systems could provide information about the configuration and topology of the network and could be considered a privacy threat.

10.6 YANG schema tree definitions[4,5]

A simplified graphical representation of the data model is used in this document. The meaning of the symbols in these diagrams is as follows:

— Brackets "[" and "]" enclose list keys.
— Abbreviations before data node names: "rw" means configuration (read-write), and "ro" means state data (read-only).
— Symbols after data node names: "?" means an optional node, "!" means a presence container, and "*" denotes a list and leaf-list.
— Parentheses enclose choice and case nodes, and case nodes are also marked with a colon (":").
— Ellipsis ("...") stand for contents of subtrees that are not shown.

10.6.1 Schema for the *ieee802-dot1ax-linkagg* YANG module

```
module: ieee802-dot1ax-linkagg
  +--rw linkagg
     +--rw agg-system* [name]
     |  +--rw name                     string
     |  +--rw actor-system?            ieee:mac-address
     |  +--rw actor-system-priority?   uint16
     +--rw key-group* [name]
        +--rw name                             string
        +--rw actor-admin-key                  uint16
        +--rw agg-system-name                  agg-system-ref
        +--rw actor-protocol-da?               ieee:mac-address
        +--rw collector-max-delay?             uint16
        +--rw actor-admin-state?               ax:lacp-state
        +--rw partner-admin-system?            ieee:mac-address
        +--rw partner-admin-system-priority?   uint16
        +--rw port-algorithm?                  identityref
        +--ro lags*                            if:interface-ref
        +--ro aggports*                        if:interface-ref
        +--rw cscd {ax:cscd}?
           +--rw admin-conv-service-map {ax:sid-map}?
           |  +--rw (method)?
           |     +--:(pattern)
           |     |  +--rw pattern?    identityref
           |     +--:(cid-list)
           |        +--rw cid-list* [cid]
           |           +--rw cid            uint16
           |           +--rw service-ids*   uint32
           +--ro admin-conv-service-digest?   binary
           +--rw admin-conv-link-map
           |  +--rw (method)?
           |     +--:(pattern)
           |     |  +--rw pattern?    identityref
           |     +--:(cid-list)
           |        +--rw cid-list* [cid]
           |           +--rw cid             uint16
           |           +--rw link-numbers*   uint16
           +--ro admin-conv-link-digest?      binary
           +--rw admin-discard-wrong-conv?    enumeration {ax:dwc}?

  augment /if:interfaces/if:interface:
    +--rw lag
       +--rw key-group-name    key-group-ref
       +--ro mac-address?      ieee:mac-address
       +--ro lacp
       |  +--ro actor-oper-key?            uint16
```

[4]*Copyright release for YANG modules*: Users of this standard may freely reproduce the YANG modules contained in this subclause so that they can be used for their intended purpose.

[5]An ASCII version of the YANG module(s) can be obtained by Web browser from the IEEE 802.1 Website at https://1.ieee802.org/yang-modules/.

```
     |  +--ro partner-system?            ieee:mac-address
     |  +--ro partner-system-priority?   uint16
     |  +--ro partner-oper-key?          uint16
     |  +--ro aggregate-or-individual?   boolean
     +--ro cscd {ax:cscd}?
        +--ro partner-port-algorithm?        identityref
        +--ro partner-conv-service-digest?   binary
        +--ro partner-conv-link-digest?      binary
        +--ro discard-wrong-conversation?    boolean
        +--ro active-lag-links*              uint16
augment /if:interfaces/if:interface/if:statistics:
  +--ro lag-stats
     +--ro unknown-protocol-frames?   yang:counter64
augment /if:interfaces/if:interface:
  +--rw aggport!
     +--rw key-group-name                 key-group-ref
     +--rw actor-port-number?             uint16
     +--rw actor-port-priority?           uint16
     +--rw partner-admin-key?             uint16
     +--rw partner-admin-port?            uint16
     +--rw partner-admin-port-priority?   uint16
     +--rw partner-admin-state?           ax:lacp-state
     +--rw wtr-time?                      uint16
     +--rw wtr-revertive?                 boolean
     +--ro lacp
     |  +--ro actor-lacp-version?             uint8
     |  +--ro actor-oper-key?                 uint16
     |  +--ro actor-oper-state?               ax:lacp-state
     |  +--ro partner-lacp-version?           uint8
     |  +--ro partner-oper-system-priority?   uint16
     |  +--ro partner-oper-system?            ieee:mac-address
     |  +--ro partner-oper-key?               uint16
     |  +--ro partner-oper-port?              uint16
     |  +--ro partner-oper-port-priority?     uint16
     |  +--ro partner-oper-state?             ax:lacp-state
     |  +--ro aggregate-or-individual?        boolean
     |  +--ro wtr-waiting?                    boolean
     +--rw cscd {ax:cscd}?
        +--rw admin-link-number?     uint16
        +--ro partner-link-number?   uint16
        +--ro link-number?           uint16
augment /if:interfaces/if:interface/if:statistics:
  +--ro aggport-stats
     +--ro lacp-pdu-rx?               yang:counter64
     +--ro marker-pdu-rx?             yang:counter64
     +--ro marker-response-pdu-rx?   yang:counter64
     +--ro unknown-rx?                yang:counter64
     +--ro illegal-rx?                yang:counter64
     +--ro lacp-pdu-tx?               yang:counter64
     +--ro marker-pdu-tx?             yang:counter64
     +--ro marker-response-pdu-tx?   yang:counter64
```

10.6.2 Schema for the *ieee802-dot1ax-drni* YANG module

```
module: ieee802-dot1ax-drni

  augment /if:interfaces/if:interface/dot1ax:lag:
    +--rw drni!
       +--rw irp-name                          if:interface-ref
       +--rw drni-aggregator-key?              uint16
       +--rw drni-aggregator-system?           ieee:mac-address
       +--rw drni-aggregator-system-priority?  uint16
       +--rw drcp-protocol-da?                 ieee:mac-address
       +--rw home-admin-irp-state?             irp-state
       +--ro home-oper-irp-state?              irp-state
       +--rw home-cscd-gateway-control?        boolean
       +--rw home-dr-client-gateway-control?   boolean
       +--rw home-gateway-algorithm?           identityref
       +--rw home-gateway-conv-service-map {ax:sid-map}?
       |  +--rw (method)?
```

```
     |      +--:(pattern)
     |      |  +--rw pattern?    identityref
     |      +--:(cid-list)
     |         +--rw cid-list* [cid]
     |            +--rw cid            uint16
     |            +--rw service-ids*   uint32
     +--ro home-gateway-conv-service-digest?   binary
     +--rw gateway-enable-mask
     |  +--rw cid-str*       string
     |  +--rw invert-mask?   boolean
     +--rw gateway-preference-mask
     |  +--rw cid-str*       string
     |  +--rw invert-mask?   boolean
     +--ro gateway-available-mask
     |  +--ro cid-str*       string
     |  +--ro invert-mask?   boolean
     +--ro neighbor
        +--ro oper-irp-state?                   irp-state
        +--ro system?                           ieee:mac-address
        +--ro system-priority?                  uint16
        +--ro drni-key?                         uint16
        +--ro aggregator-algorithm?             identityref
        +--ro aggregator-conv-service-digest?   binary
        +--ro aggregator-conv-link-digest?      binary
        +--ro partner-system-priority?          uint16
        +--ro partner-system?                   ieee:mac-address
        +--ro partner-aggregator-key?           uint16
        +--ro cscd-state?                       bits
        +--ro active-links*                     uint16
        +--ro gateway-algorithm?                identityref
        +--ro gateway-conv-service-digest?      binary
        +--ro gateway-available-mask
        |  +--ro cid-str*       string
        |  +--ro invert-mask?   boolean
        +--ro gateway-preference-mask
           +--ro cid-str*       string
           +--ro invert-mask?   boolean
  augment /if:interfaces/if:interface/if:statistics:
    +--ro drni-stats
       +--ro drcpdus-rx?   yang:counter64
       +--ro illegal-rx?   yang:counter64
       +--ro drcpdus-tx?   yang:counter64
```

10.7 YANG modules

10.7.1 The *ieee802-dot1ax-types* YANG module

```
module ieee802-dot1ax-types {
  yang-version 1.1;
  namespace "urn:ieee:params:xml:ns:yang:ieee802-dot1ax-types";
  prefix dot1ax-types;

  organization
    "IEEE 802.1 Working Group";
  contact
    "WG-URL: http://www.ieee802.org/1/
     WG-Email: stds-802-1-l@ieee.org

     Contact: IEEE 802.1 Working Group Chair
      Postal: C/O IEEE 802.1 Working Group
              IEEE Standards Association
              45 Hoes Lane
              Piscataway, NJ  08854
              USA
      E-mail: stds-802-1-chairs@ieee.org";
  description
    "Common types used within 802.1AX Link Aggregation modules.

```

```
     Copyright (C) IEEE (2024).

     This version of this YANG module is part of IEEE Std 802.1AX;
     see the standard itself for full legal notices.";

  revision 2025-02-07 {
    description
      "Published as part of IEEE Std 802.1AXdz-202y.
       The following reference statement identifies each referenced IEEE
       Standard as updated by applicable amendments.";
    reference
      "IEEE Std 802.1AX-2020 Link Aggregation:
       IEEE Std 802.1AXdz-202y.";
  }

  feature cscd {
    description
      "Conversation Sensitive Collection and Distribution (CSCD)
       is supported.";
    reference
      "5.3.2, 6.6 of IEEE Std 802.1AX";
  }

  feature dwc {
    description
      "The Discard Wrong Conversation option in CSCD is
       supported.";
    reference
      "5.3.2, 6.6 of IEEE Std 802.1AX";
  }

  feature sid-map {
    description
      "The Service ID to Conversation ID map is supported for
       use by distribution algorithms.";
  }

  feature c-vid-dist-alg {
    description
      "Supports distribution based on C-VIDs.";
  }

  feature s-vid-dist-alg {
    description
      "Supports distribution based on S-VIDs.";
  }

  feature i-sid-dist-alg {
    description
      "Supports distribution based on I-SIDs.";
  }

  feature te-sid-dist-alg {
    description
      "Supports distribution based on TE-SIDs.";
  }

  feature flow-hash-dist-alg {
    description
      "Supports distribution based on Flow Hash.";
  }

  feature basic-link-maps {
    description
      "Pre-defined admin-conv-link-map configurations to distribute
       packets on either one or two active links, with any
       remaining links available as standby.";
  }

  identity distribution-algorithm {
    description
      "Each distribution algorithm is identified by a sequence of
```

```
      4 octets, structured as shown in Figure 8-1. Distribution
      algorithm identifiers are used by network administrators to
      select between algorithms and, in Conversation-sensitive
      LACP and (if supported) Distributed Resilient Network
      Interconnect (DRNI) operation, to check whether partners
      and neighbors are using the same algorithm.

      This identity is intended to serve as base identity, not
      to be directly referenced.

      Vendor specific, combination (ex: multi-layer), and other
      customized distribution algorithms can be created as
      their own identities in their own YANG files, derived from
      this imported base type.  When the identity description
      specifies a 4 octet value, this value is used for cscd
      purposes, and included in LACPDUs. Otherwise the identity
      is used purely for local selection of a distribution
      algorithm, and the unspecified distribution algorithm is
      used for cscd purposes.";
  reference
    "8.1, 8.2 of IEEE Std 802.1AX";
}

identity unspecified {
  base distribution-algorithm;
  description
    "The 'Unspecified distribution algorithm' identifier has
     been reserved for use when the algorithm is unknown (or
     is not advertised).
     The algorithm identifier is 00-80-C2-00.";
  reference
    "Table 8-1 of IEEE Std 802.1AX";
}

identity c-vids-nomap {
  if-feature "c-vid-dist-alg";
  base distribution-algorithm;
  description
    "Distribution based on C-VIDs (8.2.1). No Service ID
     mapping table is used.
     The algorithm identifier is 00-80-C2-01.";
  reference
    "Table 8-1 of IEEE Std 802.1AX";
}

identity c-vids-map {
  if-feature "c-vid-dist-alg and sid-map";
  base distribution-algorithm;
  description
    "Distribution based on C-VIDs (8.2.1). A Service ID
     mapping table is used.
     The algorithm identifier is 00-80-C2-81.";
  reference
    "Table 8-1 of IEEE Std 802.1AX";
}

identity s-vids-nomap {
  if-feature "s-vid-dist-alg";
  base distribution-algorithm;
  description
    "Distribution based on S-VIDs (8.2.2). No Service ID
     mapping table is used.
     The algorithm identifier is 00-80-C2-02.";
  reference
    "Table 8-1 of IEEE Std 802.1AX";
}

identity s-vids-map {
  if-feature "s-vid-dist-alg and sid-map";
  base distribution-algorithm;
  description
    "Distribution based on S-VIDs (8.2.2). A Service ID
```

```
       mapping table is used.
       The algorithm identifier is 00-80-C2-82.";
    reference
      "Table 8-1 of IEEE Std 802.1AX";
  }

  identity i-sids-nomap {
    if-feature "i-sid-dist-alg";
    base distribution-algorithm;
    description
      "Distribution based on I-SIDs (8.2.3). No Service ID
       mapping table is used.
       The algorithm identifier is 00-80-C2-03.";
    reference
      "Table 8-1 of IEEE Std 802.1AX";
  }

  identity i-sids-map {
    if-feature "i-sid-dist-alg and sid-map";
    base distribution-algorithm;
    description
      "Distribution based on I-SIDs (8.2.3). A Service ID
       mapping table is used.
       The algorithm identifier is 00-80-C2-83.";
    reference
      "Table 8-1 of IEEE Std 802.1AX";
  }

  identity te-sids-nomap {
    if-feature "te-sid-dist-alg";
    base distribution-algorithm;
    description
      "Distribution based on TE-SIDs (8.2.4). No Service ID
       mapping table is used.
       The algorithm identifier is 00-80-C2-04.";
    reference
      "Table 8-1 of IEEE Std 802.1AX";
  }

  identity te-sids-map {
    if-feature "te-sid-dist-alg and sid-map";
    base distribution-algorithm;
    description
      "Distribution based on TE-SIDs (8.2.4). A Service ID
       mapping table is used.
       The algorithm identifier is 00-80-C2-84.";
    reference
      "Table 8-1 of IEEE Std 802.1AX";
  }

  identity flow-hash-nomap {
    if-feature "flow-hash-dist-alg";
    base distribution-algorithm;
    description
      "Distribution based on Flow Hash (8.2.5). No Service ID
       mapping table is used.
       The algorithm identifier is 00-80-C2-05.";
    reference
      "Table 8-1 of IEEE Std 802.1AX";
  }

  identity flow-hash-map {
    if-feature "flow-hash-dist-alg and sid-map";
    base distribution-algorithm;
    description
      "Distribution based on Flow Hash (8.2.5). A Service ID
       mapping table is used.
       The algorithm identifier is 00-80-C2-85.";
    reference
      "Table 8-1 of IEEE Std 802.1AX";
  }
```

```
identity link-map-patterns {
  description
    "Base identity for patterns filling the admin-conv-link-map.
     This identity is intended to serve as base identity, not
     to be directly referenced. Use of the identity allows vendors
     to augment the module with vendor specified patterns,";
}

identity one-plus-n {
  if-feature "basic-link-maps";
  base link-map-patterns;
  description
    "Provides active/standby behavior with one active link and
     up to 63 standby links. All packets are mapped to whichever
     link that is active in the aggregation has the lowest link
     number. If that link fails, all packets are mapped to the
     lowest link number of the remaining links that are active
     in the aggregation.
     The map consists of an identical entry for each CID, with a
     list of numbers from 1 to 64.
     Supports any number of links in the aggregation (up to 64)
     with link numbers between 1 and 64. Any links in the
     aggregation with a link number greater than 64  are not
     used.";
}

identity two-plus-n {
  if-feature "basic-link-maps";
  base link-map-patterns;
  description
    "Provides basic load-balancing on two active links with up
     to 62 standby links. All packets with an even CID value are
     mapped to the link with the lowest link number in the
     aggregation, and all packets with an odd CID value are
     mapped to the link with the highest link number in the
     aggregation.
     The map consists of an identical entry for each even CID
     containing a list of numbers from 1 to 64, and an identical
     entry for each odd CID containing a list of numbers from 64
     to 1.
     Supports any number of links in the aggregation (up to 64)
     with link numbers between 1 and 64. Any links in the
     aggregation with a link number greater than 64 are not
     used.";
}

identity service-map-patterns {
  description
    "Base identity for patterns filling the admin-conv-service-map.
     This identity is intended to serve as base identity, not
     to be directly referenced. Use of the identity allows vendors
     to augment the module with vendor specified patterns,";
}

typedef lacp-state {
  type bits {
    bit lacp-activity {
      position 1;
      description
        "Provides administrative control over when LACPDUs are
         transmitted.  A value of '1' indicates Active mode where
         LACPDUs are sent regardless of partner's lacp-activity
         value.  A value of '0' indicates Passive mode where
         LACPDUs are sent only when the partner's lacp-activity
         value is '1' (partner is in Active mode).";
    }
    bit lacp-timeout {
      position 2;
      description
        "Provides administrative control over the frequency of
         received LACPDUs.  A value of '1' indicates Short Timeout
         (so partner uses frequent transmission).  A value of '0'
```

```
          indicates Long Timeout (so partner can use infrequent
          transmission).";
      }
      bit aggregation {
        position 3;
        description
          "Provides administrative control over whether this
           Aggregation Port can be in a LAG with more than one
           member.  A value of '1' indicates the port can be
           aggregated with other ports.  A value of '0' indicates
           the port can only be a solitary link.";
      }
      bit synchronization {
        position 4;
        description
          "The Synchronization state of the MUX state machine.";
      }
      bit collecting {
        position 5;
        description
          "The Collecting state of the MUX state machine.";
      }
      bit distributing {
        position 6;
        description
          "The Distributing state of the MUX state machine.";
      }
      bit defaulted {
        position 7;
        description
          "Indicates the port is using the partner-admin values
           to select an Aggregator.";
      }
      bit expired {
        position 8;
        description
          "The Expired state of the Receive state machine.";
      }
    }
    description
      "LACP state values as transmitted in LACPDUs.
       Administrative control over the values of lacp-activity,
       lacp-timeout, aggregation, and (in partner-admin-lacp-state)
       synchronization is allowed.
       The remaining bits are read-only.";
    reference
      "6.4.1, 6.4.2.3 of IEEE Std 802.1AX";
  }

  grouping link-map {
    description
      "Specifies the contents of the admin-conv-link-map.";
    choice method {
      default "cid-list";
      description
        "Provides two ways to specify the map contents.";
      leaf pattern {
        type identityref {
          base link-map-patterns;
        }
        description
          "Use a predefined pattern to fill the map.";
      }
      list cid-list {
        key "cid";
        description
          "Data structure to map a Conversation Identifier
           (CID) to a Link Number. Each entry consists of a CID
           and a list of link numbers that can potentially be
           selected for that CID. An empty list of link-numbers
           means that no links are selected for the CID.";
        leaf cid {
```

```
          type uint16 {
            range "0..4095";
          }
          description
            "Port Conversation Identifier";
        }
        leaf-list link-numbers {
          type uint16;
          description
            "List of zero or more Link Numbers that can
             potentially be selected for distribution of frames
             with this CID.";
        }
      }
    }
  }

  grouping service-map {
    description
      "Specifies the contents of the admin-conv-service-map.";
    choice method {
      default "cid-list";
      description
        "Provides two ways to specify the map contents.";
      leaf pattern {
        type identityref {
          base service-map-patterns;
        }
        description
          "Use a predefined pattern to fill the map.";
      }
      list cid-list {
        key "cid";
        description
          "Data structure to map service identifiers to
           conversation identifiers.  Each entry consists of a
           Conversation ID (CID) and a list of zero or more
           Service Identifiers (SIDs) that map to it. An empty
           list of SIDs means there are no SIDs that map to this
           CID, and results in the same behavior as not having an
           entry for this CID.";
        leaf cid {
          type uint16 {
            range "0..4095";
          }
          description
            "Port Conversation Identifier";
        }
        leaf-list service-ids {
          type uint32;
          description
            "List of zero or more SIDs that map to the CID.";
        }
      }
    }
  }
}
```

10.7.2 The *ieee802-dot1ax-linkagg* YANG module

```
module ieee802-dot1ax-linkagg {
  yang-version 1.1;
  namespace "urn:ieee:params:xml:ns:yang:ieee802-dot1ax-linkagg";
  prefix dot1ax;

  import ieee802-dot1ax-types {
    prefix ax;
  }
  import ieee802-types {
```

```
    prefix ieee;
  }
  import ietf-yang-types {
    prefix yang;
  }
  import ietf-interfaces {
    prefix if;
  }
  import iana-if-type {
    prefix ianaif;
  }

  organization
    "IEEE 802.1 Working Group";
  contact
    "WG-URL: http://www.ieee802.org/1/
     WG-Email: stds-802-1-l@ieee.org

     Contact: IEEE 802.1 Working Group Chair
      Postal: C/O IEEE 802.1 Working Group
              IEEE Standards Association
              45 Hoes Lane
              Piscataway, NJ  08854
              USA
      E-mail: stds-802-1-chairs@ieee.org";
  description
    "This YANG module describes the configuration model for Link
     Aggregation, as specified in IEEE Std 802.1AX, including Link
     Aggregation Control Protocol (LACP) and Conversation Sensitive
     Collection and Distribution.

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     This version of this YANG module is part of IEEE Std 802.1AX;
     see the standard itself for full legal notices.";

  revision 2025-02-07 {
    description
      "Published as part of IEEE Std 802.1AXdz-202y.
       The following reference statement identifies each referenced IEEE
       Standard as updated by applicable amendments.";
    reference
      "IEEE Std 802.1AX-2020 Link Aggregation:
       IEEE Std 802.1AXdz-202y.";
  }

  typedef key-group-ref {
    type leafref {
      path "/dot1ax:linkagg/dot1ax:key-group/dot1ax:name";
    }
    description
      "This type is used by aggregators and aggregation ports to
       reference an entry in the key-group list.";
  }

  typedef agg-system-ref {
    type leafref {
      path "/dot1ax:linkagg/dot1ax:agg-system/dot1ax:name";
    }
    description
      "This type is used in the key-group list entries to
       reference an entry in the agg-system list.";
  }

  augment "/if:interfaces/if:interface" {
    when
      "derived-from-or-self(if:type,'ianaif:ieee8023adLag') or "
    + "derived-from-or-self(if:type,'ianaif:ieee8021axDrni')" {
      description
        "Applies to interfaces representing a LAG or
         (if supported) DRNI Intra-Relay Port.";
    }
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    description
      "Augment Interface with Aggregator parameters.";
    container lag {
      description
        "Contains the configuration and status information that
         allows an instance of an Aggregator to be managed.";
      leaf key-group-name {
        type key-group-ref;
        mandatory true;
        description
          "Specifies the entry in the link-aggregation key-group
           list to which this aggregator is assigned.";
      }
      leaf mac-address {
        type ieee:mac-address;
        config false;
        description
          "The MAC address assigned to the Aggregator.";
        reference
          "7.3.1.1.9 of IEEE Std 802.1AX";
      }
      container lacp {
        config false;
        description
          "Contains aggregator LACP operational data.";
        leaf actor-oper-key {
          type uint16;
          description
            "The current operational value of the Key for the
             Aggregator. The meaning of particular Key
             values is of local significance.";
          reference
            "7.3.1.1.8 of IEEE Std 802.1AX";
        }
        leaf partner-system {
          type ieee:mac-address;
          description
            "The value of the MAC address portion of the
             System Identifier for the current
             protocol partner of this Aggregator. A value
             of zero indicates that there is no known partner.
             If the aggregation is manually configured, this
             System identifer value is assigned by the
             local System.";
          reference
            "7.3.1.1.10 of IEEE Std 802.1AX";
        }
        leaf partner-system-priority {
          type uint16;
          description
            "Indicates the priority value portion of the current
             Partner's System Identifier. If the aggregation is
             manually configured, this System Priority value is
             assigned by the local System.";
          reference
            "7.3.1.1.11 of IEEE Std 802.1AX";
        }
        leaf partner-oper-key {
          type uint16;
          description
            "The current operational value of the Key for the
             Aggregators current protocol Partner. If the
             aggregation is manually configured, this Key value
             is assigned by the local System.";
          reference
            "7.3.1.1.12 of IEEE Std 802.1AX";
        }
        leaf aggregate-or-individual {
          type boolean;
          description
            "Indicates whether the Aggregator represents an
             Aggregate (TRUE) or an Individual link (FALSE).";
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        reference
          "7.3.1.1.6 of IEEE Std 802.1AX";
      }
    }
    container cscd {
      if-feature "ax:cscd";
      config false;
      description
        "Aggregator parameters obtained by the operation of LACP
         supporting CSCD.";
      leaf partner-port-algorithm {
        type identityref {
          base ax:distribution-algorithm;
        }
        description
          "Operational value of the distribution algorithm in
           use by the LACP Partner.";
        reference
          "7.3.1.1.40 of IEEE Std 802.1AX";
      }
      leaf partner-conv-service-digest {
        type binary;
        description
          "The MD5 Digest of the admin-conv-service-map in use
           by the LACP Partner.";
        reference
          "7.3.1.1.42, 6.6.3.1 of IEEE Std 802.1AX";
      }
      leaf partner-conv-link-digest {
        type binary;
        description
          "The MD5 Digest of the admin-conv-link-map in use
           by the LACP Partner.";
        reference
          "7.3.1.1.41, 6.6.3.1 of IEEE Std 802.1AX";
      }
      leaf discard-wrong-conversation {
        type boolean;
        description
          "The operational value that determines whether an
           Aggregator discards a frame that is collected
           from an Aggregation Port that is different from the
           Aggregation Port to which the Aggregator would
           distribute a frame with the same Port Conversation
           ID.";
        reference
          "7.3.1.1.37, 6.6 of IEEE Std 802.1AX";
      }
      leaf-list active-lag-links {
        type uint16;
        description
          "A list, possibly empty, of the operational
           link-number of each Aggregation Port active
           (i.e. Collecting) on this Aggregator.";
        reference
          "7.3.1.1.43 of IEEE Std 802.1AX";
      }
    }
  }
}

augment "/if:interfaces/if:interface/if:statistics" {
  when '../dot1ax:lag' {
    description
      "Applies to aggregators.";
  }
  description
    "Augment interface statistics with aggregator statistics.";
  container lag-stats {
    config false;
    description
      "Contains the set of stats associated with the
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        Aggregator.";
      leaf unknown-protocol-frames {
        type yang:counter64;
        description
          "A count of data frames discarded on reception by all
           ports that are (or have been) members of the
           aggregation, due to the detection of an unknown Slow
           Protocols PDU (7.3.3.1.5)";
        reference
          "7.3.1.1.29 of IEEE Std 802.1AX";
      }
    }
  }

  augment "/if:interfaces/if:interface" {
    description
      "Augment interface model with Aggregation port
       configuration nodes.";
    container aggport {
      presence
        "When present, this interface supports Link Aggregation";
      description
        "Contains Aggregation Port configuration related nodes,
         which provides the basic management controls necessary
         to allow an instance of an Aggregation Port to be managed,
         for the purposes of Link Aggregation.";
      leaf key-group-name {
        type key-group-ref;
        mandatory true;
        description
          "Specifies the entry in the link-aggregation key-group
           list to which this aggregation-port is assigned.";
      }
      leaf actor-port-number {
        type uint16 {
          range "1..65535";
        }
        description
          "The port number assigned to the Aggregation Port.
           The port number is communicated in LACPDUs as the
           Actor_Port.
           This leaf provides the ability to administratively
           override the initial value provided by the system.";
        reference
          "7.3.2.1.14, 6.4.6 of IEEE Std 802.1AX";
      }
      leaf actor-port-priority {
        type uint16;
        default "0x8000";
        description
          "The priority value assigned to this Aggregation Port.";
        reference
          "7.3.2.1.15, 6.4.6 of IEEE Std 802.1AX";
      }
      leaf partner-admin-key {
        type uint16 {
          range "1..65535";
        }
        description
          "The current administrative value of the Key for the
           protocol Partner. The assigned value is used, along
           with the value of port-partner-admin-system-priority,
           partner-admin-system, partner-admin-port, and
           partner-admin-port-priority, in order to achieve
           manually configured aggregation.
           The default value is the initial value of actor-port-
           number provided by the system.";
        reference
          "7.3.2.1.10 of IEEE Std 802.1AX";
      }
      leaf partner-admin-port {
        type uint16 {
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        range "1..65535";
      }
      description
        "The current administrative value of the port number for
         the protocol Partner. The assigned value is used, along
         with the value of partner-admin-system-priority,
         partner-admin-system, port-partner-admin-key, and
         partner-admin-port-priority, in order to achieve
         manually configured aggregation.
         The default value is the initial value of actor-port-
         number provided by the system.";
      reference
        "7.3.2.1.16 of IEEE Std 802.1AX";
    }
    leaf partner-admin-port-priority {
      type uint16;
      default "0";
      description
        "The current administrative value of the port priority
         for the protocol Partner. The assigned value is used,
         along with the value of partner-admin-system-priority,
         partner-admin-system, partner-admin-key, and
         partner-admin-port, in order to achieve manually
         configured aggregation.";
      reference
        "7.3.2.1.18 of IEEE Std 802.1AX";
    }
    leaf partner-admin-state {
      type ax:lacp-state {
        bit lacp-activity {
          position 1;
          description
            "When the LACP_Activity state is set to '0', the
             transmission of LACPDUs is controlled by the
             actor-admin-state.LACP_Activity.";
        }
        bit lacp-timeout {
          position 2;
          description
            "When the LACP_Timeout is set to '0', LACPDUs
             are transmitted at the Slow_Periodic_Time.";
        }
        bit aggregation {
          position 3;
          description
            "When the Aggregation state is set to '0', this
             port cannot be aggregated with any other port.";
        }
        bit synchronization {
          position 4;
          description
            "When the Synchronization state is set to '0', this
             port cannot become active.";
        }
      }
      default "synchronization";
      description
        "Provides administrative control over the partner's
         LACP_Activity, LACP_Timeout, Aggregation, and
         Synchronization state when the partner's information is
         unknown (i.e. no LACPDUs are received from the partner).";
      reference
        "7.3.2.1.22, 6.4.1, 6.4.2.3, 6.4.6 of IEEE Std 802.1AX";
    }
    leaf wtr-time {
      type uint16 {
        range "0..32767";
      }
      default "0";
      description
        "The wait-to-restore (WTR) period, in seconds, that
         needs to elapse between an Aggregation Port on a LAG
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          coming up (Port_Operational becoming TRUE) and being
          permitted to become active (transmitting and
          receiving frames) on the LAG.";
      reference
        "7.3.2.1.30 of IEEE Std 802.1AX";
    }
    leaf wtr-revertive {
      type boolean;
      default "true";
      description
        "Controls revertive or non-revertive mode of operation.
         When TRUE, the Aggregation Port can become active as
         soon as the wait-to-restore timer expires regardless of
         the state of other links in the LAG.
         When FALSE, the Aggregation Port cannot become active
         unless there are no other links that can become active
         in the LAG. The default value is TRUE.";
      reference
        "7.3.2.1.31 of IEEE Std 802.1AX";
    }
    container lacp {
      config false;
      description
        "Contains Aggregation port LACP operational related
         nodes.";
      leaf actor-lacp-version {
        type uint8;
        description
          "The version number transmitted in LACPDUs on this
           Aggregation Port";
        reference
          "7.3.2.1.33 of IEEE Std 802.1AX";
      }
      leaf actor-oper-key {
        type uint16;
        description
          "The current operational value of the Key for the
           Aggregation Port. The meaning of particular Key values
           is of local significance.";
        reference
          "7.3.2.1.5 of IEEE Std 802.1AX";
      }
      leaf actor-oper-state {
        type ax:lacp-state;
        description
          "The operational value of the Actor_State as
           transmitted in LACPDUs.";
        reference
          "7.3.2.1.21, 6.4.1, 6.4.2.3, 6.4.6 of IEEE Std 802.1AX";
      }
      leaf partner-lacp-version {
        type uint8;
        description
          "The version number in the LACPDU most recently
           received on this Aggregation Port.";
        reference
          "7.3.2.1.34 of IEEE Std 802.1AX";
      }
      leaf partner-oper-system-priority {
        type uint16;
        description
          "Indicates the operational value of priority associated
           with the Partners System ID. The value of this
           attribute can contain the manually configured value
           carried in partner-admin-system-priority if there is
           no protocol Partner.";
        reference
          "7.3.2.1.7 of IEEE Std 802.1AX";
      }
      leaf partner-oper-system {
        type ieee:mac-address;
        description
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          "Represents the MAC address portion of the Aggregation
           Port's current protocol partner's System Identifier.
           A value of zero indicates that there is no known
           protocol Partner.
           The value of this attribute can contain the manually
           configured value carried in partner-admin-system if
           there is no protocol Partner.";
        reference
          "7.3.2.1.9 of IEEE Std 802.1AX";
      }
      leaf partner-oper-key {
        type uint16;
        description
          "The current operational value of the Key for the
           protocol Partner. The value of this attribute can
           contain the manually configured value carried in
           partner-admin-key if there is no protocol Partner.";
        reference
          "7.3.2.1.11 of IEEE Std 802.1AX";
      }
      leaf partner-oper-port {
        type uint16;
        description
          "The operational port number assigned by the
           Aggregation Port's protocol Partner. The value of this
           attribute can contain the administratively configured
           value carried in partner-admin-port if there is no
           protocol Partner.";
        reference
          "7.3.2.1.17 of IEEE Std 802.1AX";
      }
      leaf partner-oper-port-priority {
        type uint16;
        description
          "The operational priority value assigned by the
           Aggregation Port's protocol Partner. The value of this
           attribute can contain the administratively configured
           value carried in partner-admin-port-priority if there
           is no protocol Partner.";
        reference
          "7.3.2.1.19 of IEEE Std 802.1AX";
      }
      leaf partner-oper-state {
        type ax:lacp-state;
        description
          "The operational value of the partner's LACP state
           derived from received LACPDUs or, when Defaulted is
           true, from the partner-admin-state.";
        reference
          "7.3.2.1.23, 6.4.1, 6.4.2.3, 6.4.6 of IEEE Std 802.1AX";
      }
      leaf aggregate-or-individual {
        type boolean;
        description
          "When true indicates the Aggregation Port can join a
           LAG consisting of multiple Aggregation Ports.
           When false, indicates that the Aggregation Port can
           only operate as an Solitary link because the
           Aggregation bit is false in either
           actor-oper-port-state or partner-oper-port-state.";
        reference
          "7.3.2.1.24 of IEEE Std 802.1AX";
      }
      leaf wtr-waiting {
        type boolean;
        description
          "Indicates the Aggregation Port is inhibited from
           becoming active for an interval (determined by
           wtr-time) after becoming operational or while
           non-revertive operation is being enforced by the
           Selection Logic.";
        reference
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          "7.3.2.32 of IEEE Std 802.1AX";
      }
    }
    container cscd {
      if-feature "ax:cscd";
      description
        "Aggregation port parameters for support of CSCD.";
      leaf admin-link-number {
        type uint16;
        description
          "The Link_Number value for the Aggregation Port,
           that is unique among all Aggregation Ports in the same
           key group. This leaf provides the ability to
           administratively override the initial value provided
           by the system. A value of 0 is allowed, but
           results in no frames being distributed to this
           Aggregation Port.
           The default value is the initial value of actor-port-
           number provided by the system.";
        reference
          "7.3.2.1.27, 6.6.3.2 of IEEE Std 802.1AX";
      }
      leaf partner-link-number {
        type uint16;
        config false;
        description
          "The last received value of the Partner_Link_Number,
           or zero if the Aggregation Port is using default
           values for the Partner or the Partner LACP Version
           is 1.";
        reference
          "7.3.2.1.29 of IEEE Std 802.1AX";
      }
      leaf link-number {
        type uint16;
        config false;
        description
          "The operational link number for this Aggregation Port.
           The value is either the same as the admin-link-number,
           or the corresponding value of the LACP partner.";
        reference
          "7.3.2.1.28 of IEEE Std 802.1AX";
      }
    }
  }
}

augment "/if:interfaces/if:interface/if:statistics" {
  when '../dot1ax:aggport' {
    description
      "Applies to aggregation ports.";
  }
  description
    "Augment interface statistics with aggport statistics.";
  container aggport-stats {
    config false;
    description
      "Contains stats associated with the Aggregation Port.";
    leaf lacp-pdu-rx {
      type yang:counter64;
      description
        "The number of valid LACPDUs received on this
         Aggregation Port.";
      reference
        "7.3.3.1.2 of IEEE Std 802.1AX";
    }
    leaf marker-pdu-rx {
      type yang:counter64;
      description
        "The number of valid Marker PDUs received on this
         Aggregation Port.";
      reference
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          "7.3.3.1.3 of IEEE Std 802.1AX";
      }
      leaf marker-response-pdu-rx {
        type yang:counter64;
        description
          "The number of valid Marker Response PDUs received on
           this Aggregation Port.";
        reference
          "7.3.3.1.4 of IEEE Std 802.1AX";
      }
      leaf unknown-rx {
        type yang:counter64;
        description
          "The number of frames received that either:
           a) Carry the Slow Protocols Ethernet Type value,
           but contain an unknown PDU, or
           b) Are addressed to the Slow Protocols group MAC
           Address, but do not carry the Slow Protocols
           Ethernet Type.";
        reference
          "7.3.3.1.5 of IEEE Std 802.1AX";
      }
      leaf illegal-rx {
        type yang:counter64;
        description
          "The number of frames received that carry the Slow
           Protocols Ethernet Type value, but contain a badly
           formed PDU or an illegal value of Protocol Subtype.";
        reference
          "7.3.3.1.6 of IEEE Std 802.1AX";
      }
      leaf lacp-pdu-tx {
        type yang:counter64;
        description
          "The number of LACPDUs transmitted on this
           Aggregation Port.";
        reference
          "7.3.3.1.7 of IEEE Std 802.1AX";
      }
      leaf marker-pdu-tx {
        type yang:counter64;
        description
          "The number of Marker PDUs transmitted on this
           Aggregation Port.";
        reference
          "7.3.3.1.8 of IEEE Std 802.1AX";
      }
      leaf marker-response-pdu-tx {
        type yang:counter64;
        description
          "The number of Marker Response PDUs transmitted on
           this Aggregation Port.";
        reference
          "7.3.3.1.9 of IEEE Std 802.1AX";
      }
    }
  }

  container linkagg {
    description
      "Link Aaggregation System specific configuration nodes.";
    list agg-system {
      key "name";
      description
        "List of aggregation systems.";
      leaf name {
        type string;
        description
          "Name for the aggregation system.";
      }
      leaf actor-system {
        type ieee:mac-address;
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        description
          "The part of the System Identifier that is a globally
           unique MAC address.  This leaf provides the ability to
           administratively override the initial value provided
           by the system.";
        reference
          "7.3.1.1.4, 7.3.2.1.3 of IEEE Std 802.1AX";
      }
      leaf actor-system-priority {
        type uint16;
        default "0x8000";
        description
          "The part of the System Identifier that is the
           priority of the system.";
        reference
          "7.3.1.1.5, 7.3.2.1.2  of IEEE Std 802.1AX";
      }
    }
    list key-group {
      key "name";
      unique "actor-admin-key";
      description
        "List of key groups.  A key group is the set of aggregators
         and aggregation ports that share the same system priority,
         system identifier, and aggregation key, and therefore can
         potentially form a Link Aggregation Group.  Each entry in
         the key group list contains the parameters common to all
         aggregation ports and/or aggregators in the key group.";
      reference
        "6.3.5, 6.4.12 of IEEE Std 802.1AX";
      leaf name {
        type string;
        description
          "Name for the key group.";
      }
      leaf actor-admin-key {
        type uint16 {
          range "1..65535";
        }
        mandatory true;
        description
          "The administrative value of the key used by the
           Aggregators and Aggregation Ports in this key-group.";
        reference
          "7.3.1.1.7, 7.3.2.1.4 of IEEE Std 802.1AX";
      }
      leaf agg-system-name {
        type agg-system-ref;
        mandatory true;
        description
          "Specifies the aggregation system for this key
           group.";
      }
      leaf actor-protocol-da {
        type ieee:mac-address;
        must ". = '01-80-c2-00-00-00' or . = '01-80-C2-00-00-00' or
              . = '01-80-c2-00-00-02' or . = '01-80-C2-00-00-02' or
              . = '01-80-c2-00-00-03' or . = '01-80-C2-00-00-03'" {
          error-message "Invalid protocol address";
        }
        default "01-80-c2-00-00-02";
        description
          "A 6-octet read-write MAC Address value specifying the DA
           to be used when sending Link Aggregation Control and
           Marker PDUs.  Valid addresses are the Nearest Customer
           Bridge, Slow_Protocols_Multicast, and Nearest non-TPMR
           Bridge group addresses.  The default value
           is the Slow_Protocols_Multicast address.";
        reference
          "7.3.2.2.1, 6.2.10.2 of IEEE Std 802.1AX";
      }
      leaf collector-max-delay {
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        type uint16;
        description
          "Specifies the maximum delay, in tens of microseconds,
           between receiving a frame from an Aggregator Port, and
           either delivering the frame to the Aggregator Client or
           discarding the frame. A value of zero means the delay
           is less than the minimum increment (< 10us).
           This leaf provides the ability to administratively
           override the initial value provided by the system.";
        reference
          "7.3.1.1.32, 6.2.3.1.1, B.3 of IEEE Std 802.1AX";
      }
      leaf actor-admin-state {
        type ax:lacp-state {
          bit lacp-activity {
            position 1;
            description
              "Setting the LACP_Activity state to '0' means that the
               transmission of LACPDUs is controlled by the value of
               partner-oper-state.LACP_Activity.";
          }
          bit lacp-timeout {
            position 2;
            description
              "Setting the LACP_Timeout to '0' means that actor uses
               the Long_Timeout value, allowing the partner to transmit
               LACPDUs at the Slow_Periodic_Time.";
          }
          bit aggregation {
            position 3;
            description
              "Setting the Aggregation state to '0' prevents this
               port from being aggregated with any other ports.";
          }
        }
        default "lacp-activity lacp-timeout aggregation";
        description
          "Provides administrative control over the values of the
           LACP_Activity, LACP_Timeout, and Aggregation state.";
        reference
          "7.3.2.1.20, 6.4.1, 6.4.2.3, 6.4.6 of IEEE Std 802.1AX";
      }
      leaf partner-admin-system {
        type ieee:mac-address;
        default "00-00-00-00-00-00";
        description
          "The administrative value of the MAC address portion of
           the Partner's System Identifier.
           The assigned value is used, along with the value of
           port-partner-admin-system, partner-admin-key,
           partner-admin-port, and partner-admin-port-priority,
           to achieve administratively configured Link
           Aggregation Groups with a partner that does not run
           LACP.";
        reference
          "7.3.2.1.8 of IEEE Std 802.1AX";
      }
      leaf partner-admin-system-priority {
        type uint16;
        default "0";
        description
          "The administrative value of priority portion of the
           the Partner's System Identifier. The assigned
           value is used, along with the value of
           port-partner-admin-system, partner-admin-key,
           partner-admin-port, and partner-admin-port-priority,
           to achieve administratively configured Link
           Aggregation Groups with a partner that does not run
           LACP.";
        reference
          "7.3.2.1.6 of IEEE Std 802.1AX";
      }
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    leaf port-algorithm {
      type identityref {
        base ax:distribution-algorithm;
      }
      default "ax:unspecified";
      description
        "Identifies the algorithm used by the Aggregator to
         assign frames to a Port Conversation ID.  Default is
         the value for an unspecified distribution algorithm.
         When the identity description specifies a 4 octet value,
         this value will be used for cscd purposes, and included
         in LACPDUs. Otherwise this leaf is used purely for
         local selection of a distribution algorithm, and the
         unspecified distribution algorithm is used for cscd
         purposes.";
      reference
        "7.3.1.1.33, Table 8-1 of IEEE Std 802.1AX";
    }
    leaf-list lags {
      type if:interface-ref;
      config false;
      description
        "A list of the if:name of aggregators assigned to this
         key group.";
      reference
        "linkagg:key-group";
    }
    leaf-list aggports {
      type if:interface-ref;
      config false;
      description
        "A list of the if:name of aggregation ports assigned to
         this key group.";
      reference
        "linkagg:key-group";
    }
    container cscd {
      if-feature "ax:cscd";
      description
        "Contains CSCD parameters that need to be consistent for
         all aggregation ports and aggregators in the key group.";
      container admin-conv-service-map {
        if-feature "ax:sid-map";
        description
          "Data structure to map service identifiers to
           conversation identifiers.  Each entry consists of a
           Conversation ID (CID) and a list of zero or more
           Service Identifiers (SIDs) that map to it. An empty
           list of SIDs means there are no SIDs that map to this
           CID, and results in the same behavior as not having an
           entry for this CID.";
        reference
          "7.3.1.1.36, 6.6.3.1 of IEEE Std 802.1AX";
        uses ax:service-map;
      }
      leaf admin-conv-service-digest {
        type binary;
        config false;
        description
          "The MD5 Digest of the admin-conv-service-map. The
           value is NULL (AAAAAAAAAAAAAAAAAAAAAA== in base64) when
           the distribution algorithm specified
           specified by agg-port-algorithm does not use the
           admin-conv-service-map.";
        reference
          "7.3.1.1.39, 6.6.3.1 of IEEE Std 802.1AX";
      }
      container admin-conv-link-map {
        description
          "Data structure to map a Conversation Identifier
           (CID) to a Link Number. Each entry consists of a CID
           and a list of link numbers that can potentially be
```

```
            selected for that CID. An empty list of link-numbers
            means that no links are selected for the CID.";
        reference
          "7.3.1.1.34, 6.6.3.1 of IEEE Std 802.1AX";
        uses ax:link-map;
      }
      leaf admin-conv-link-digest {
        type binary;
        config false;
        description
          "The MD5 Digest of the admin-conv-link-map. The value
           value is NULL (AAAAAAAAAAAAAAAAAAAAAA== in base64) when
           the distribution algorithm specified
           agg-port-algorithm does not use the
           admin-conv-link-map.";
        reference
          "7.3.1.1.38, 6.6.3.1 of IEEE Std 802.1AX";
      }
      leaf admin-discard-wrong-conv {
        if-feature "ax:dwc";
        type enumeration {
          enum force-true {
            value 1;
            description
              "Indicates that an Aggregator discards a
               frame that is collected from an Aggregation Port
               that is different from the Aggregation Port to
               which the Aggregator would distribute a frame
               with the same Port Conversation ID.";
          }
          enum force-false {
            value 2;
            description
              "Indicates that an Aggregator does not discard
               a frame that is collected from an Aggregation Port
               that is different from the Aggregation Port to
               which the Aggregator would distribute a frame with
               the same Port Conversation ID. This is the behavior
               of the Aggregator when DWC is not supported";
          }
          enum auto {
            value 3;
            description
              "Indicates that the Aggregator behaves as
               if the value was force-true only when the actor
               and partner agree on the algorithms (other than
               unspecified) and mapping tables used to map frames
               to Aggregation Ports, and behaves as if the value
               was force-false otherwise.";
          }
        }
        default "force-false";
        description
          "Indicates whether an Aggregator discards a
           frame that is collected from an Aggregation Port
           that is different from the Aggregation Port to which
           the Aggregator would distribute a frame with the
           same Port Conversation ID.";
        reference
          "7.3.1.1.35, 6.6 of IEEE Std 802.1AX";
      }
    }
  }
 }
}
```

10.7.3 The *ieee802-dot1ax-drni* YANG module

```
module ieee802-dot1ax-drni {
```

```
yang-version 1.1;
namespace "urn:ieee:params:xml:ns:yang:ieee802-dot1ax-drni";
prefix dot1ax-drni;

import ieee802-dot1ax-types {
  prefix ax;
}
import ieee802-dot1ax-linkagg {
  prefix dot1ax;
}
import ieee802-types {
  prefix ieee;
}
import ietf-yang-types {
  prefix yang;
}
import ietf-interfaces {
  prefix if;
}

organization
  "IEEE 802.1 Working Group";
contact
  "WG-URL: http://www.ieee802.org/1/
   WG-Email: stds-802-1-l@ieee.org

   Contact: IEEE 802.1 Working Group Chair
    Postal: C/O IEEE 802.1 Working Group
            IEEE Standards Association
            45 Hoes Lane
            Piscataway, NJ  08854
            USA
    E-mail: stds-802-1-chairs@ieee.org";
description
  "This YANG module describes the configuration model for a
   Distributed Resilient Network Interface (DRNI) as specified
   in 802.1AX.

   Copyright (C) IEEE (2024).

   This version of this YANG module is part of IEEE Std 802.1AX;
   see the standard itself for full legal notices.";

revision 2025-02-07 {
  description
    "Published as part of IEEE Std 802.1AXdz-202y.
     The following reference statement identifies each referenced IEEE
     Standard as updated by applicable amendments.";
  reference
    "IEEE Std 802.1AX-2020 Link Aggregation:
     IEEE Std 802.1AXdz-202y.";
}

typedef irp-state {
  type bits {
    bit reserved-1 {
      position 1;
      description
        "Reserved for future use. It is set to 0 on
         transmit and ignored on receipt.";
    }
    bit reserved-2 {
      position 2;
      description
        "Reserved for future use. It is set to 0 on
         transmit and ignored on receipt.";
    }
    bit short-timeout {
      position 3;
      description
        "The Short_Timeout flag indicates the Timeout control value
         in use by the DRCP Receive machine on this IRP. Short Timeout
```

```
          is encoded as a 1; Long Timeout is encoded as a 0.";
      }
      bit synchronization {
        position 4;
        description
          "When the Sync flag is TRUE (1), the DRCP Receive machine has
           determined the Neighbor DRNI System has a compatible
           configuration for forming a DRNI.";
      }
      bit irc-data {
        position 5;
        description
          "When the IRC_Data flag is TRUE (1), the transfer of Up
           and Down frames is permitted on the IRC.";
      }
      bit drni {
        position 6;
        description
          "The DRNI flag is TRUE (1) when this DRNI System is paired
           with another DRNI System and FALSE (0) otherwise.";
      }
      bit defaulted {
        position 7;
        description
          "When the Defaulted flag is TRUE (1), the DRCP Receive machine
           is using default operational Neighbor information.
           When FALSE (0), the operational Neighbor information
           in use has been received in a DRCPDU.";
      }
      bit expired {
        position 8;
        description
          "When the Expired flag is TRUE (1), the DRCP Receive machine
           is in the EXPIRED state.";
      }
    }
    description
      "IRP state values as transmitted in DRCPDUs.
       Administrative control over the values of Short_Timeout
       and IRC_Data is allowed. The remaining bits are read-only.";
    reference
      "7.4.1.1.24, 9.6.2.3, Figure 9-13 of IEEE Std 802.1AX";
  }

  grouping drni-mask {
    description
      "Specifies the contents of a bit mask indexed by CID.";
    leaf-list cid-str {
      type string {
        pattern '([0-9]{1}|[1-9]{1}[0-9]{1,2}|[1-3]{1}[0-9]{3}'
              + '|40[0-8]{1}[0-9]{1}|409[0-5]{1})'
              + '((-([0-9]{1}|[1-9]{1}[0-9]{1,2}|[1-3]{1}[0-9]{3}'
              + '|40[0-8]{1}[0-9]{1}|409[0-5]{1})'
              + '(%[0-9]{1,4})?)?)'
              + '(,'
              + '([0-9]{1}|[1-9]{1}[0-9]{1,2}|[1-3]{1}[0-9]{3}'
              + '|40[0-8]{1}[0-9]{1}|409[0-5]{1})'
              + '((-([0-9]{1}|[1-9]{1}[0-9]{1,2}|[1-3]{1}[0-9]{3}'
              + '|40[0-8]{1}[0-9]{1}|409[0-5]{1})'
              + '(%[0-9]{1,4})?)?)'
              + ')*';
      }
      default "0-4095";
      description
        "A comma separated list of single CID values (x),
         ranges (x-y), or ranges with an increment (x-y%z).
         For example, the string '0,10-13,100-110%4' sets the
         bits in the mask corresponding to CIDs
         0,10,11,12,13,100,104,108.
         Some special cases are:
          'x-y%z' where x+z>y is the same as 'x'.
          'x-y%0' is the same as 'x'.
```

```
         'x-y%1' is the same as 'x-y'.
         'x-y%2' sets the bit for all even CIDs in the range
         when x is even, and for all odd CIDs when x is odd.
        Specified as a list of strings, so that long strings
        can be broken into several shorter strings.";
   }
   leaf invert-mask {
     type boolean;
     default "false";
     description
       "When true the specified mask is inverted:
        each zero replaced with a one,
        and each one replaced with a zero.";
   }
 }

 augment "/if:interfaces/if:interface/dot1ax:lag" {
   description
     "Augmentation parameters only for Aggregators with
      DRNI enabled.";
   container drni {
     presence "When present, this Aggregator is enabled for DRNI";
     description
       "Aggregator parameters to support a Distributed
        Resilient Network Interface";
     leaf irp-name {
       type if:interface-ref;
       mandatory true;
       description
         "Interface Name (if:name) of the Port supporting the
          Intra Relay Port (IRP) of this DRNI Gateway.";
       reference
         "7.4.1.1.23 of IEEE Std 802.1AX";
     }
     leaf drni-aggregator-key {
       type uint16 {
         range "1..65535";
       }
       description
         "The Aggregator Key value to be used by the Aggregator
          supporting this DRNI Gateway (and the Aggregation Ports
          assigned to this DRNI Gateway) when paired with a
          neighbor DRNI System via the IRC. The default value is
          the Aggregator’s Actor_Admin_Aggregator_Key value.";
       reference
         "7.4.1.1.15 of IEEE Std 802.1AX";
     }
     leaf drni-aggregator-system {
       type ieee:mac-address;
       default "00-00-00-00-00-00";
       description
         "The Aggregator System value to be used by the
          Aggregator supporting this DRNI Gateway (and the
          Aggregation Ports assigned to this DRNI Gateway)
          when paired with a neighbor DRNI System via the
          Intra-Relay Connection (IRC).";
       reference
         "7.4.1.1.13 of IEEE Std 802.1AX";
     }
     leaf drni-aggregator-system-priority {
       type uint16;
       default "0";
       description
         "The Aggregator System Priority value to be used by the
          Aggregator supporting this DRNI Gateway (and the
          Aggregation Ports assigned to this DRNI Gateway) when
          paired with a neighbor DRNI System via the IRC.";
       reference
         "7.4.1.1.14 of IEEE Std 802.1AX";
     }
     leaf drcp-protocol-da {
       type ieee:mac-address;
```

```
        must ". = '01-80-c2-00-00-00' or . = '01-80-C2-00-00-00' or
              . = '01-80-c2-00-00-0e' or . = '01-80-C2-00-00-0E' or
              . = '01-80-c2-00-00-03' or . = '01-80-C2-00-00-03'" {
          error-message "Invalid protocol address";
        }
        default "01-80-c2-00-00-03";
        description
          "A 6-octet read-write MAC Address value specifying the
           Destination Address for Distributed Relay Control PDUs
           transmitted on the Intra-Relay Port. Valid addresses are
           the Nearest Customer Bridge, Nearest Bridge, and
           Nearest non-TPMR Bridge group addresses.  The default
           value is the Nearest Non-TPMR Bridge group
           address.";
        reference
          "7.4.1.1.12, 9.6.1.1 of IEEE Std 802.1AX";
      }
      leaf home-admin-irp-state {
        type irp-state {
          bit short-timeout {
            position 3;
            description
              "The Short_Timeout flag indicates the Timeout control value
               in use by the DRCP Receive machine on this IRP. Short Timeout
               is encoded as a 1; Long Timeout is encoded as a 0.";
          }
          bit irc-data {
            position 5;
            description
              "When the IRC_Data flag is TRUE (1), the transfer of Up
               and Down frames is permitted on the IRC.";
          }
        }
        default "short-timeout irc-data";
        description
          "Provides administrative control over the values of the
           Short_Timeout and IRC-Data state.";
        reference
          "7.4.1.1.24, 9.6.2.3, Figure 9-13 of IEEE Std 802.1AX";
      }
      leaf home-oper-irp-state {
        type irp-state;
        config false;
        description
          "A string of 8 bits, corresponding to the current
           operational value of IRP_State as transmitted in
           DRCPDUs.";
        reference
          "7.4.1.1.25, 9.6.2.3, Figure 9-13 of IEEE Std 802.1AX";
      }
      leaf home-cscd-gateway-control {
        type boolean;
        default "true";
        description
          "When TRUE, allows the DRNI Gateway Port selection to
           be based on the CSCD parameters that control the
           Aggregator Port selection.";
        reference
          "7.4.1.1.16 of IEEE Std 802.1AX";
      }
      leaf home-dr-client-gateway-control {
        type boolean;
        default "true";
        description
          "When TRUE, allows the Distributed Relay Client to
           determine whether to forward frames through the DRNI
           Gateway Port.";
        reference
          "7.4.1.1.17 of IEEE Std 802.1AX";
      }
      leaf home-gateway-algorithm {
        type identityref {
```

```
        base ax:distribution-algorithm;
      }
      default "ax:unspecified";
      description
        "Identifies the algorithm used by the DRNI Gateway to
         assign frames to a Gateway Conversation ID. 8.2 provides
         the IEEE 802.1 OUI (00-80-C2) Gateway Algorithm
         encodings.  Default is the value for an unspecified
         distribution algorithm.";
      reference
        "7.4.1.1.6 of IEEE Std 802.1AX";
    }
    container home-gateway-conv-service-map {
      if-feature "ax:sid-map";
      description
        "Data structure to map service identifiers to
         conversation identifiers. Each entry consists of a
         Conversation ID (CID) and a list of zero or more Service
         Identifiers (SIDs) that map to it. Frames with Service
         IDs not contained in the map are not mapped to any
         Gateway Conversation ID and are discarded.";
      reference
        "7.4.1.1.20, 6.6.3.1 of IEEE Std 802.1AX";
      uses ax:service-map;
    }
    leaf home-gateway-conv-service-digest {
      type binary;
      config false;
      description
        "The MD5 Digest of the home-gateway-conv-service-map. The
         value is NULL (AAAAAAAAAAAAAAAAAAAAAA== in base64) when
         the distribution algorithm specified
         by agg-port-algorithm does not use the
         home-gateway-conv-service-map.";
      reference
        "7.4.1.1.21 of IEEE Std 802.1AX";
    }
    container gateway-enable-mask {
      description
        "A vector of Boolean values, with one value for each
         possible Gateway Conversation ID. A 1 indicates that
         frames associated with that Gateway Conversation ID
         are allowed to pass through this Gateway Port, and a
         0 indicates that such frames are not allowed to pass.
         Default value is all bits set to 1.";
      reference
        "7.4.1.1.18, 9.5.3.5, 9.6.5 of IEEE Std 802.1AX";
      uses drni-mask;
    }
    container gateway-preference-mask {
      description
        "A vector of Boolean values, with one value for each
         possible Gateway Conversation ID. A 1 indicates that
         this Gateway Port is the preferred Gateway when both
         DRNI Gateways have the Gateway Conversation ID enabled
         in the gateway-available-mask, and a 0 indicates that
         it is not preferred.
         Default value is all bits set to 1.";
      reference
        "7.4.1.1.19, 9.5.3.5, 9.6.5 of IEEE Std 802.1AX";
      uses drni-mask;
    }
    container gateway-available-mask {
      config false;
      description
        "A vector of Boolean values, with one value for each
         possible Gateway Conversation ID. A 1 indicates that
         this Gateway Port is eligible to be selected to pass
         that Gateway Conversation ID, and a 0 indicates that
         it is not eligible.";
      reference
        "7.4.1.1.22, 9.5.3.5, 9.6.5 of IEEE Std 802.1AX";
```

```
        uses drni-mask;
      }
      container neighbor {
        config false;
        description
          "Operational values for the DRNI neighbor obtained
           from DRCPDUs.";
        leaf oper-irp-state {
          type irp-state;
          description
            "A string of 8 bits, corresponding to the current
             operational value of IRP_State as transmitted in
             DRCPDUs.";
          reference
            "7.4.1.1.25, 9.6.2.3, Figure 9-13 of IEEE Std 802.1AX";
        }
        leaf system {
          type ieee:mac-address;
          description
            "The MAC Address portion of the System Identifier of
             the Neighbor DRNI System (connected via the
             Intra-Relay Port). ";
          reference
            "7.4.1.1.29 of IEEE Std 802.1AX";
        }
        leaf system-priority {
          type uint16;
          description
            "The priority portion of the System Identifier of the
             Neighbor DRNI System (connected via the Intra-Relay
             Port).";
          reference
            "7.4.1.1.30 of IEEE Std 802.1AX";
        }
        leaf drni-key {
          type uint16;
          description
            "The DRNI key value received from the Neighbor DRNI
             System (connected via the IntraRelay Port).";
          reference
            "7.4.1.1.31 of IEEE Std 802.1AX";
        }
        leaf aggregator-algorithm {
          type identityref {
            base ax:distribution-algorithm;
          }
          description
            "The Port algorithm used by the Neighbor Aggregator to
             assign frames to Port Conversation IDs.";
          reference
            "7.4.1.1.33 of IEEE Std 802.1AX";
        }
        leaf aggregator-conv-service-digest {
          type binary;
          description
            "The MD5 Digest of the Neighbor Aggregator's
             Admin_Conv_Service_Map. Obtained from the Home
             Aggregator State TLV last received from the Neighbor
             DRNI System.";
          reference
            "7.4.1.1.34 of IEEE Std 802.1AX";
        }
        leaf aggregator-conv-link-digest {
          type binary;
          description
            "The MD5 Digest of the Neighbor Aggregator's
             Admin_Conv_Link_Map. Obtained from the Home Aggregator
             State TLV (9.6.2.4) last received from the Neighbor
             DRNI System.";
          reference
            "7.4.1.1.35 of IEEE Std 802.1AX";
        }
```

```
        leaf partner-system-priority {
          type uint16;
          description
            "The priority portion of the System Identifier of the
             Neighbor Aggregator's Partner.";
          reference
            "7.4.1.1.36 of IEEE Std 802.1AX";
        }
        leaf partner-system {
          type ieee:mac-address;
          description
            "The MAC Address portion of the System Identifier of
             the Neighbor Aggregator's Partner.";
          reference
            "7.4.1.1.37 of IEEE Std 802.1AX";
        }
        leaf partner-aggregator-key {
          type uint16;
          description
            "The operational key value of the Neighbor
             Aggregator's Partner.";
          reference
            "7.4.1.1.38 of IEEE Std 802.1AX";
        }
        leaf cscd-state {
          type bits {
            bit reserved-1 {
              position 0;
              description
                "Bit 1 is reserved for future use. It is set to 0
                 and ignored on receipt.";
            }
            bit reserved-2 {
              position 1;
              description
                "Bit 2 is reserved for future use. It is set to 0
                 and ignored on receipt.";
            }
            bit reserved-3 {
              position 2;
              description
                "Bit 3 is reserved for future use. It is set to 0
                 and ignored on receipt.";
            }
            bit cscd_gateway_control {
              position 3;
              description
                "CSCD_Gateway_Control is encoded in bit 4. When
                 this flag is TRUE, the DRNI Gateway is configured
                 to minimize forwarding data frames on the IRC by
                 selecting the DRNI Gateway and Aggregator Ports
                 for forwarding any given Conversation ID to be in
                 the same DRNI System.";
            }
            bit discard_wrong_conversation {
              position 4;
              description
                "Discard_Wrong_Conversation is encoded in bit 5.
                 The Aggregator's Discard_Wrong_Conversation
                 value.";
            }
            bit differ_conv_link_digests {
              position 5;
              description
                "Differ_Conv_Link_Digests is encoded in bit 6.
                 This flag is TRUE when the Aggregator's
                 Actor_Conv_Link_Digest matches the Aggregator's
                 Partner_Conv_Link_Digest.";
            }
            bit differ_conv_service_digests {
              position 6;
              description
```

```
              "Differ_Conv_Service_Digests is encoded in bit 7.
               This flag is TRUE when the Aggregator's
               Actor_Conv_Service_Digest matches the Aggregator's
               Partner_Conv_Service_Digest.";
          }
          bit differ_port_algorithms {
            position 7;
            description
              "Differ_Port_Algorithms is encoded in bit 8. The
               Aggregator's differPortAlgorithms flag is TRUE
               when the Aggregator's Actor_Port_Algorithm matches
               the Aggregator's Partner_Port_Algorithm.";
          }
        }
        description
          "8 bits, corresponding to the Aggregator_CSCD_State
           in the Neighbor_Aggregator_State variable. The first
           three bits (the least significant bits of CSCD_State)
           are reserved; the fourth bit corresponds to the
           Neighbor's value for Home_Admin_CSCD_Gateway_Control;
           the fifth bit corresponds to the Neighbor Aggregator's
           operational value for Discard_Wrong_Conversation; and
           the sixth, seventh, and eighth bits correspond to the
           Neighbor Aggregator's operational value for
           differConvLinkDigests, differConvServiceDigests, and
           differPortAlgorithms, respectively, (the most
           significant bits of CSCD_State).";
        reference
          "7.4.1.1.39 of IEEE Std 802.1AX";
      }
      leaf-list active-links {
        type uint16;
        description
          "A list of the operational Link_Numbers of Aggregation
           Ports that are currently active (i.e., collecting) on
           the Neighbor's Aggregator. An empty list indicates that
           there are no Aggregation Ports active. Each integer
           value in the list carries an aAggPortOperLinkNumber
           attribute value.";
        reference
          "7.4.1.1.40 of IEEE Std 802.1AX";
      }
      leaf gateway-algorithm {
        type identityref {
          base ax:distribution-algorithm;
        }
        description
          "The gateway algorithm used by the Neighbor DRNI
           Gateway to assign frames to Gateway Conversation IDs.";
        reference
          "7.4.1.1.41 of IEEE Std 802.1AX";
      }
      leaf gateway-conv-service-digest {
        type binary;
        description
          "The MD5 Digest of the Neighbor DRNI Gateway's
           the Home_Admin_Gateway_Conv_Service_Map. Obtained
           from Gateway_Conv_Service_Digest in the
           Neighbor_Gateway_State TLV last received from the
           Neighbor DRNI System.";
        reference
          "7.4.1.1.42 of IEEE Std 802.1AX";
      }
      container gateway-available-mask {
        description
          "A vector of Boolean values, with one value for each
           possible Gateway Conversation ID. A 1 indicates that
           the Neighbor DRNI Gateway Port is eligible to be
           selected to pass that Gateway Conversation ID, and
           a 0 indicates that it is not eligible.";
        reference
          "7.4.1.1.43 of IEEE Std 802.1AX";
```

```
          uses drni-mask;
        }
        container gateway-preference-mask {
          description
            "A vector of Boolean values, with one value for each
             possible Gateway Conversation ID. A 1 indicates that
             the Neighbor DRNI Gateway Port is the preferred
             Gateway when both DRNI Gateways have the Gateway
             Conversation ID enabled in the gateway-available-mask,
             and a 0 indicates that it is not preferred.";
          reference
            "7.4.1.1.44 of IEEE Std 802.1AX";
          uses drni-mask;
        }
      }
    }
  }

  augment "/if:interfaces/if:interface/if:statistics" {
    when '../dot1ax:lag/dot1ax-drni:drni' {
      description
        "Applies to aggregators with DRNI present.";
    }
    description
      "Augment interface statistics with DRNI statistics.";
    container drni-stats {
      description
        "Contains DRNI specific statistics.";
      leaf drcpdus-rx {
        type yang:counter64;
        description
          "The number of valid DRCPDUs received on this
           Intra-Relay Port.";
        reference
          "7.4.1.1.45 of IEEE Std 802.1AX";
      }
      leaf illegal-rx {
        type yang:counter64;
        description
          "The number of frames received on this Intra-Relay
           Port that carry the DRCP EtherType value,
           but contain a badly formed PDU.";
        reference
          "7.4.1.1.46, 9.6.1.4 of IEEE Std 802.1AX";
      }
      leaf drcpdus-tx {
        type yang:counter64;
        description
          "The number of valid DRCPDUs transmitted on this
           Intra-Relay Port.";
        reference
          "7.4.1.1.47 of IEEE Std 802.1AX";
      }
    }
  }
}
```

Annex A

(normative)

Protocol implementation conformance statement (PICS) proforma[6]

Insert the following rows at the end of the table in A.2.1

A.2.1 Major capabilities/options

Item	Feature	Status	References	Support
YANG	Does the implementation support management operations using YANG modules?	O	10.6	Yes [ ] No [ ]
YANG modules	Is the ieee802-dot1ax-types module supported?	YANG:M	10.7.1	N/A [ ] Yes [ ]
	Is the ieee802-dot1ax-linkagg module supported?	YANG:M	10.7.2	N/A [ ] Yes [ ]
	Is the ieee802-dot1ax-drni module supported?	DRNI AND YANG:M	10.7.3	N/A [ ] Yes [ ]

[6] *Copyright release for PICS proformas:* Users of this standard may freely reproduce the PICS proforma in this subclause so that it can be used for its intended purpose and may further publish the completed PICS.

Annex G

(informative)

Bibliography

Insert the following two bibliography entries after [B8] in Annex G:

[B9] IETF RFC 7803, Changing the Registration Policy for the NETCONF Capability URNs Registry, February 2016.

[B10] IETF RFC 8040, RESTCONF Protocol, January 2017.

[B11] OMG Unified Modeling Language (OMG UML), Version 2.5, March 2015.[7]

[7]OMG documents are available from the Object Management Group (https://www.omg.org/).